

Kinematics

Class 11 Physics | Nine Education IIT Academy

Name: _____ Date: _____

Section A: Equations of Motion

1. A cyclist accelerates uniformly from rest at 2 m/s^2 for 6 s. Find the final velocity and the distance covered.
2. A car moving at 20 m/s decelerates uniformly and comes to rest after covering 100 m. Find the deceleration and the time taken.
3. A skateboarder moves with $u = 4 \text{ m/s}$ and uniform acceleration 1.5 m/s^2 . Find the velocity at $t = 4 \text{ s}$ and the distance covered.
4. An e-scooter accelerates uniformly from 3 m/s to 15 m/s in 4 s. Find the acceleration and the distance covered.
5. A metro train travels at 22 m/s for 15 s, then decelerates uniformly at 2 m/s^2 until it stops. Find the total distance covered during the entire motion.
6. A motorcycle starts from rest and accelerates uniformly at 5 m/s^2 for 8 s. Find its final velocity and the distance covered.
7. A body has initial velocity 25 m/s and uniform deceleration 5 m/s^2 . Find the time it takes to stop and the distance covered before stopping.
8. A particle has $u = 8 \text{ m/s}$, $v = 28 \text{ m/s}$, and uniform acceleration 4 m/s^2 . Find the time taken and the distance covered.
9. A cyclist decelerates uniformly from 14 m/s and comes to rest after 7 s. Find the deceleration and the distance covered.
11. A car travels at 15 m/s for 8 s, then accelerates uniformly at 3 m/s^2 for the next 5 s. Find the total distance covered.
12. A body starts from rest and covers 50 m in 5 s under uniform acceleration. Find the acceleration and the final velocity.
13. A truck moving at 18 m/s decelerates uniformly and comes to rest within 60 m. Find the deceleration and the time taken.
14. A drone accelerates uniformly from rest at 3 m/s^2 for 7 s. Find its final velocity and the vertical distance covered.
15. A bike moving at 10 m/s accelerates uniformly at 2 m/s^2 for 5 s. Find its final velocity and the distance covered.
16. A particle has $u = 10 \text{ m/s}$, $v = 40 \text{ m/s}$, and covers a distance of 125 m under uniform acceleration. Find the acceleration and the time taken.
17. A car accelerates uniformly from rest to 25 m/s in 10 s, travels at this speed for 20 s, and then decelerates uniformly to rest in 5 s. Find the total distance covered and the average speed for the whole journey.
18. A particle moves with an initial velocity of 15 m/s and a uniform deceleration of 3 m/s^2 , eventually reversing direction. Find the total distance travelled by the particle in the first 8 seconds.
19. A bullet moving at 200 m/s strikes a wooden plank of thickness 10 cm and emerges with a velocity of 100 m/s, losing speed at a uniform rate. Find the retardation. If an identical plank is placed immediately behind the first, determine whether the bullet will emerge from it - and if not, how much further into it the bullet travels before stopping.

10. A runner increases speed uniformly from 3 m/s to 9 m/s in 3 s. Find the acceleration and the distance covered.

20. Two cars, 250 m apart, start moving towards each other at the same instant. Car A starts from rest and accelerates uniformly at 2 m/s^2 . Car B moves with a uniform velocity of 15 m/s. Find the time after which the two cars meet, and the distance covered by each car at that instant.

Section B: Displacement in the Nth Second

21. A body starts from rest with uniform acceleration 2 m/s^2 . Find the displacement in the 3rd second.

22. A particle starts from rest with uniform acceleration 10 m/s^2 . Find (a) the distance covered in 3 s, and (b) the distance covered in the 3rd second. State the units of each answer clearly.

23. A body starts from rest with uniform acceleration 8 m/s^2 . Find the displacement in the 6th second.

24. A particle has $u = 3 \text{ m/s}$ and uniform acceleration 2 m/s^2 . Find its displacement in the 5th second.

25. A particle has $u = 6 \text{ m/s}$ and uniform acceleration 3 m/s^2 . Find its displacement in the 4th second.

26. A body starts from rest with uniform acceleration 9 m/s^2 . Find the ratio of distances covered in the 1st, 2nd, and 3rd seconds.

27. A car moving with uniform acceleration covers 8 m in the 2nd second and 12 m in the 4th second. Find its initial velocity and acceleration.

28. A body starts from rest with uniform acceleration and covers a total distance of 400 m in 10 s. Find the distance it covers in the 10th second.

29. A particle has an initial velocity of 3 m/s and a uniform acceleration of 6 m/s^2 . It covers 24 m during a certain second of its motion. Find which second this is.

30. A particle starts from rest and moves with uniform acceleration 3.5 m/s^2 . Show that the distances it covers in successive seconds ($S_1, S_2, S_3, \dots$) form an arithmetic progression, and state the common difference. Hence find the sum of the distances covered in the first 6 seconds using the AP sum formula, and verify this matches the value obtained directly from $s = (1/2)at^2$.

Integration

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Section A: Power Rule

1. $\int x^6 dx$

2. $\int x^{-4} dx$

3. $\int \sqrt{x} dx$

4. $\int 1 / \sqrt{x} dx$

6. $\int 4x^3 dx$

7. $\int -3x^2 dx$

8. $\int 7 dx$

9. $\int -5 dx$

5. $\int x^{2/3} dx$

Section B: Standard Trigonometric Integrals

10. $\int \sin x dx$

13. $\int \operatorname{cosec}^2 x dx$

11. $\int \cos x dx$

14. $\int \sec x \tan x dx$

12. $\int \sec^2 x dx$

15. $\int \operatorname{cosec} x \cot x dx$

Section C: Exponential & Logarithmic Form

16. $\int e^x dx$

18. $\int 1/x dx$

17. $\int 2^x dx$

Section D: Sums of Direct-Formula Terms

19. $\int 3x^5 - 2x^3 + x dx$

25. $\int 2 \sin x - 3 \cos x dx$

20. $\int 4 - 3x + x^2 dx$

26. $\int \sec^2 x + 5 dx$

21. $\int x^3 + x^2 + x + 1 dx$

27. $\int e^x + x dx$

22. $\int 6x^2 - 4x + 9 dx$

28. $\int 5x^4 - 3\sqrt{x} dx$

23. $\int 1/x^2 + 3x dx$

29. $\int x^{-2} - x^{-3} dx$

24. $\int \sqrt{x} + 1/\sqrt{x} dx$

30. $\int 2/x + 3e^x dx$

Kinematics — Answer Key

1. $v = 12 \text{ m/s}$, $s = 36 \text{ m}$

16. $a = 6 \text{ m/s}^2$, $t = 5 \text{ s}$

2. deceleration = 2 m/s^2 , $t = 10 \text{ s}$

17. total distance = 687.5 m , average speed $\approx 19.64 \text{ m/s}$

3. $v = 10 \text{ m/s}$, $s = 28 \text{ m}$

18. total distance = 51 m

4. $a = 3 \text{ m/s}^2$, $s = 36 \text{ m}$

19. retardation = $1.5 \times 10^5 \text{ m/s}^2$; bullet does NOT emerge from the second plank; travels $\approx 3.3 \text{ cm}$ further before stopping

5. $s = 451 \text{ m}$

20. $t = 10 \text{ s}$; Car A covers 100 m , Car B covers 150 m

6. $v = 40 \text{ m/s}$, $s = 160 \text{ m}$

21. 5 m

7. $t = 5 \text{ s}$, $s = 62.5 \text{ m}$

22. (a) 45 m , (b) 25 m

8. $t = 5 \text{ s}$, $s = 90 \text{ m}$

23. 44 m

9. deceleration = 2 m/s^2 , $s = 49 \text{ m}$

24. 12 m

10. $a = 2 \text{ m/s}^2$, $s = 18 \text{ m}$

25. 16.5 m

11. $s = 232.5 \text{ m}$

26. $1 : 3 : 5$

12. $a = 4 \text{ m/s}^2$, $v = 20 \text{ m/s}$

13. deceleration $= 2.7 \text{ m/s}^2$, $t \approx 6.67 \text{ s}$

14. $v = 21 \text{ m/s}$, $s = 73.5 \text{ m}$

15. $v = 20 \text{ m/s}$, $s = 75 \text{ m}$

27. $u = 5 \text{ m/s}$, $a = 2 \text{ m/s}^2$

28. 76 m

29. 4th second

30. common difference $= 3.5 \text{ m/s}^2$; sum of first 6 s $= 63 \text{ m}$ (matches $s = (1/2)at^2 = 63 \text{ m}$)

Integration — Answer Key

1. $x^7/7 + C$

2. $-1/(3x^3) + C$

3. $(2/3)x^{3/2} + C$

4. $2\sqrt{x} + C$

5. $(3/5)x^{5/3} + C$

6. $x^4 + C$

7. $-x^3 + C$

8. $7x + C$

9. $-5x + C$

10. $-\cos x + C$

11. $\sin x + C$

12. $\tan x + C$

13. $-\cot x + C$

14. $\sec x + C$

15. $-\operatorname{cosec} x + C$

16. $e^x + C$

17. $2^x / \ln 2 + C$

18. $\ln|x| + C$

19. $x^6/2 - x^4/2 + x^2/2 + C$

20. $4x - (3/2)x^2 + x^3/3 + C$

21. $x^4/4 + x^3/3 + x^2/2 + x + C$

22. $2x^3 - 2x^2 + 9x + C$

23. $-1/x + (3/2)x^2 + C$

24. $(2/3)x^{3/2} + 2\sqrt{x} + C$

25. $-2\cos x - 3\sin x + C$

26. $\tan x + 5x + C$

27. $e^x + x^2/2 + C$

28. $x^5 - 2x^{3/2} + C$

29. $-1/x + 1/(2x^2) + C$

30. $2 \ln|x| + 3e^x + C$